

JANVI P. MADHANI
jmadhan1@jhu.edu | janvimadhani.github.io

PROFILE

Sixth and final year Physics and Astronomy graduate student and Ph.D. candidate at Johns Hopkins University with demonstrated experience in instrumentation, space telescope survey design, data analysis, supercomputing, and hydrodynamic simulations. Strongly committed to social justice in STEM through long-haul mentoring projects in collaboration with Baltimore Area and NYC public universities. Interested in developing theoretical models, numerical simulations, and pattern recognition projects related to Cosmology, Galaxy Evolution, Earth & Atmospheric physics, and Response to Climate Change.

GRADUATE RESEARCH EXPERIENCE:

August 2020 - Present: Graduate Research Assistant, Johns Hopkins University and City University of New York. Advisor(s): C. Welker (Primary advisor, CUNY), C. Norman (JHU, advisor)

Graduate Research Assistant | Johns Hopkins University, MD
advised by Dr. Charlotte Welker and Dr. Colin Norman
since Aug 2020
responsibilities: research, teaching
research topics: hydrodynamic simulations, pattern detection in large-scale structures, JWST and CLASS (CMB) survey design, preparation and/or calibration

LSST/NRAO Graduate Research Assistant | City University of New York, NYC
College of Technology, NY
advised by Dr. Charlotte Welker.
March 2023 -Dec 2024
responsibilities: research, undergraduate mentoring, outreach
research topics: planes of satellite galaxies, pattern detection in large-scale structures, cosmological hydrodynamic simulations, data analysis

Planes of Satellite Galaxies | Johns Hopkins University, advisor: C. Welker.
Skills: fluid dynamics modeling and programming, data analysis, pattern recognition, supercomputing

- Resolved a long-standing tension between satellite galaxies' kinematics and the standard model of cosmology, using cosmological, hydrodynamic simulations.
- developed infrastructural and analysis software in Fortran and Python to extract dark matter and gas filaments, identify planar structures among galaxies and filaments and detect alignments patterns in cosmological simulations.
- Developed a method to adapt and calibrate a structure finding algorithm to identification of local gas streams around galaxies as well as large scale cosmic filaments.
- Conducted an in-depth analysis that evidenced strong correlations between satellite dynamics and gas dynamics on larger scale (vorticity) (1st author publication submitted.)

Since June 2021

Graduate Research Assistant | Space Telescope Science Institute, MD
advised by Dr. Charlotte Welker and Dr. Susan Kassin.
June 2021- Jan 2023
responsibilities: research, teaching
research topics: pattern detection in large-scale cosmic structures, JWST galaxy survey design and preparation

Galaxies in Cosmic Filaments at $z = 3$ with JWST | Space Telescope Science Institute, advisors: C. Welker, S. Kassin.

Skills: space telescope survey planning, spectroscopy, calibration

Since June 2021

- Core co-I of a successful JWST Cycle 2 proposal to detect the influence of cosmic filaments on the spin galaxies at $z \approx 3$ with deep spectroscopy. (PI: Kassin)
- Demonstrated the possibility to detect cosmic filaments on CEERS galaxy data, developed, tested and calibrated a filament detection pipeline.
- Estimated the minimal specifications of the planned spectroscopic survey for optimal detection of galaxy spin alignments with filaments.

Cosmology Large Angular Scale Surveyor (CLASS) | Johns Hopkins University, advisor: T. Marriage.

Skills: atmospheric noise analysis, instrumentation, signal processing, data reduction

Aug 2020 - May 2021

- Conducted data selection and analysis with CLASS, a large angular scale cosmic microwave background (CMB) experiment searching for gravitational B-modes
- Developed the post-demodulated cuts pipeline to clean data for CMB mapping.
- Characterized the atmospheric polarization signal at large angular scales.
- Tested and assembled cryogenic instrumentation.

UNDERGRADUATE RESEARCH EXPERIENCE: _____

Allegheny Observatory Parallax Catalogue | University of Pittsburgh, advisor: D. Turnshek.

Skills: database creation and management

Sept 2019 - Jul 2020

- Automated the creation of catalogues of parallaxes for the Allegheny Observatory, cross-matched them with the Yale Parallax Catalogue, SIMBAD and AO historic photographic plates.
- Created a searchable database of stars (specific to the observed Allegheny Observatory parallaxes).

Testing a Theoretical Cosmological Model | University of Pittsburgh, advisor: A. Kosowsky.

Skills: Analytical cosmology modeling

Jan 2019 - Apr 2020

- Investigated the effects of an exotic cosmological model on the power spectrum of the Cosmic Microwave Background using both analytical and computational models.

NASA Eclipse Ballooning Project | University of Pittsburgh, advisor: D. Turnshek.

Skills: programming, atmosphere analysis, instrumentation, mechanical engineering

Sept 2016 - Nov 2019

- Studied “shadow bands”, the bands of light and dark racing across the surface of the earth right before totality, with a high-altitude balloon during the 2017 total solar eclipse.

- Designed and created a lab shadow band simulator as well as balloon and ground-based photo-diode circuits.

- Analyzed eclipse data in search of shadow bands in the upper atmosphere. (led to 1st author publication)

Searching for Fast Radio Bursts with the ACT Telescope | University of Pittsburgh, advisor: A. Kosowsky.

Skills: programming, structure identification

Jan 2018 - May 2019

- Automated the identification of Fast Radio Bursts (FRBs) candidate in the Atacama Cosmology Telescope (ACT) data.

- Successfully developed a pipeline to differentiate cosmic rays from glitch data.

EDUCATION

Johns Hopkins University, Baltimore, Maryland, USA

since Aug 2020

M.A. (2023) Ph.D. Candidate in Physics & Astronomy

advisor: C. Welker, C. Norman

University of Pittsburgh, Pittsburgh, Pennsylvania, USA

2015 - 2019

B.S. in Physics & Astronomy, *magna cum laude*

REFEREED AND SUBMITTED PUBLICATIONS

1. **Madhani, J. M.**, Welker, C., Nair, S., Gallego, D., Feliciano, L., Pichon, C., Olsen, C., Dubois, Y., Kaviraj, S., Kraljic, K., “*Zippers and Twisters: Planes of satellite galaxies emerge from whirling and shocking gas streams in the cosmic web (38 pages)*” **ApJ**, Submitted, March 2025. [Read here!](#)
2. **Madhani, J. M.**, Chu, G. E., Gomez, C.V., Bartel, S., Clark, R.J., Coban, L. W., Hartman, M., Potosky, E.M., Rao, S.M., Bonato, M., Turnshek, D. A., “*Observation of Eclipse Shadow Bands Using High Altitude Balloon and Ground-Based Photodiode Arrays,*” **JASTP**, 2020, 211, 105420. [Read here!](#)

PUBLICATIONS IN PREPARATION

1. **Madhani, J. M.**, Welker, C., Nair, S., “*Non-Linear Scales of Filament Detection in Gas Rich Cosmic Environments within NewHorizon*”, In prep.
2. **Madhani, J. M.**, Welker, C., “*Understanding Lopsideness of Satellite Distributions in NewHorizon*”, In prep.

TEACHING AND MENTORING EXPERIENCE

CUNY City Tech, since 2023	LSST Research Mentor <i>Activities: co-advising on research projects, lead of collaborative code development, graduate application counselling, conference chaperone</i> • Undergraduate Students: Lianys Feliciano, Daniel Gallego • Graduate Students: Manuel Ramirez, Sneha Nair, Sophia Miskiewicz
STScI/ Johns Hopkins 2022	Astro Scholars Instructor <i>This NSF program offers a week-long boot camp in research, programming, and astrophysics, followed by tailored mentoring and monthly meetings for college students from underrepresented groups in the Baltimore and D.C. area.</i> • Participated in group meetings and mentoring • Conducted reviews and interviews as a member of the admission committee • Developed and taught the week-long introductory Python course (10 students).
Johns Hopkins 2021-2022	Research Mentorship <i>Activities: co-advising on astronomy topics, technical support, co-working sessions</i> • High School Student, Catherine Franklin • Undergraduate Student, Angela Wroblewski
Johns Hopkins since 2020	JHU Teaching Assistant • Stellar Astrophysics (<i>Fall 2025</i>), taught by Dr. Meredith MacGregor. • Electricity and Magnetism I (<i>Spring 2025</i>), taught by Dr. Danielle Speller. • Stellar Structure and Evolution (<i>Grad, Fall 2023</i>), taught by Dr. Kevin Schlaufman. • General Physics for Physical Sciences (<i>Summer 2023</i>), taught by Dr. Reid Mumford. • General Physics for Bio Majors I (<i>Fall 2020</i>), taught by Dr. Tobias Marriage. • General Physics Lab I (<i>Fall 2020</i>), taught by Dr. Reid Mumford.
U. of Pittsburgh 2018 - 2020	Undergraduate Teaching • Undergraduate Tutor for the Physics Department • Undergraduate Teaching Assistant for Quantum Mechanics I (2018)

COMPUTATIONAL SKILLS

	Computer Languages • Fluent: Python, bash, git, Matlab. • Familiar: FORTRAN, C
	Software & Packages • numpy, scipy, matplotlib, astropy, pandas, L^AT_EX, Microsoft Office, RAMSES, DisPerSE, SUNSET, MUSIC, SolidWorks

GRANTS AND FUNDING AWARDS

ChOIR Collab: New Data that Challenge Underlying Assumptions in Early Galaxy Evolution

PI, Awarded July 2025, \$400

Travel Grants

IAU S379: Dynamical Masses of Local Group Galaxies

PI, Awarded by Leibniz Institut fur Astrophysik March 2023, €1100

AAS FAMOUS Travel Grant

PI, Awarded January 2022, \$1000

Research

JWST 4291: Galaxy angular momentum alignment with filaments at $z \sim 3$: The effect of large scale structure on galaxies

Co-Investigator, Cycle 2 - 2023 (PI:Kassin)

Space Telescope Science Institute Director's Discretionary Research Fund

Awarded 2021 and 2022, \$9000 for Summer and \$60,000 for a full year

Co-Investigator (PI: Kassin, Welker)

NASA - Pennsylvania Space Grant Consortium Scholarship

PI, Awarded 2017, 2018, 2019, Cumulative total of \$15,000

OUTREACH

Jan 17, 2020

Panel Speaker for CUWiP, University of Pittsburgh

- Spoke on panels to share my experiences as an early career researcher.

*Oct 2016
2017, 2018, 2019*

Allegheny Observatory Open House, University of Pittsburgh

- Volunteered at the annual Allegheny Observatory open house giving talks some years and doing physics demonstrations other years for the general public.

Jan 30, 2019

Science Research Panel Speaker – Internship Week, University of Pittsburgh

March 2018

Public Lecture - Allegheny Observatory Open House, University of Pittsburgh

PROFESSIONAL SERVICE

May 2-4, 2022

Congressional Visits Day, Washington D.C.

- Met with members of Congress to advocate for science policy and research funding on behalf of the AAS.

IN THE PRESS

May 20, 2025

Why do dwarf galaxies line up? ‘Zippers’ and ‘twisters’ in the early universe may solve a galactic mystery, *space.com*

[Click to read!](#)

May 12, 2025

Episode 11: Whirling planes, wandering black holes and alien supernovae, *starxiv podcast*

- Recording can be found [here!](#)

June 14, 2022

Press Conference at AAS 240, Pasadena, CA

- Recording can be found [here!](#)

Aug 21, 2017 **Live Radio Interview with P.J. Maloney of KQV AM 1410, Pittsburgh**

Aug 16, 2017 **Shadow Bandits Ready for Eclipse Day, University of Pittsburgh**
• [Click to read!](#)

July 15, 2017 **It's up, up, and away..., University of Pittsburgh**
• [Click to read!](#)

INVITED TALKS AND WORKSHOPS

July 30, 2025 **Plenary Talk: "Astronomy and Militarism"** | ChOIR Conference, New Data that Challenges Underlying Assumptions in Early Galaxy Evolution (*Winter Harbor, ME*)

June 4, 2025 **Workshop: "Extracting Gas Filaments in NewHorizon: Using DisPerSE on Gas and Galaxies"** | Gotham Web Lab, City University of New York (*Brooklyn, NY*)

Apr 19, 2023 **"The Formation of Planes of Satellite Galaxies"** | University of Connecticut Galaxy Seminar (*virtual*).

March 16, 2018 **Plenary Talk: "Eclipse Shadow Bands"** | Duquesne University's Summer Research Symposium (*Pittsburgh, PA*)

Sept 22, 2017 **Plenary Talk: "Observations of Eclipse Shadow Bands"** | Allegheny Observatory Open House (*Pittsburgh, PA*)

CONTRIBUTED TALKS & LECTURES

October 9, 2025 **"The Cosmic Origins of Planes of Satellites"** | IAU S403: The Hidden Beauty of the Galactic Outskirts (*University of Cordoba, Cordoba, Spain*)

July 24, 2023 **"The Formation of Planes of Satellite Galaxies"** | Galactic Frontiers: Dwarf Galaxies in the Local Volume and Beyond (*Center for Computational Astrophysics, NYC*)

March 24, 2023 **"The Formation of Planes of Satellite Galaxies"** | IAU S379: Dynamical Masses of Local Group Galaxies (*Potsdam, Germany*)

Feb 6, 2023 **"Conflict Resolved: Planes of Satellites Found in Simulations"** | IAU S377: Early Disk-Galaxy Formation from JWST to the Milky Way (*Kuala Lumpur, Malaysia*)

Sept 23, 2022 **"Planes of Satellite Galaxies"** | Galaxy Group Meeting at the Center for Computational Astrophysics (*virtual*)

June 13, 2022 **"The Formation of Planes of Satellite Galaxies"** | 240th Meeting of the AAS (*Pasadena, CA*)

POSTERS

July 29-Aug 2, 2024 **Planes of Satellite Galaxies Trace Environmental Whirling Gas Streams in the Cosmic Web** | Small Galaxies, Cosmic Questions (*Durham, UK*)

March 20-24, 2023 **The Formation of Planes of Satellite Galaxies** | IAU S379: Dynamical Masses of Local Group Galaxies (*Potsdam, Germany*)
• Won "Most Innovative and Creative Poster" for which I was awarded a contributed talk slot.

<i>May 23-27, 2022</i>	Planes of Satellite Galaxies in New Horizon IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution (<i>Granada, Spain</i>)
<i>Jan 4, 2019</i>	Observations of Eclipse Shadow Bands 235th Meeting of the AAS (<i>Honolulu, HI</i>)
<i>Oct 13, 2016</i>	NASA Eclipse Ballooning Project White House Frontiers Astronomy Night (<i>Pittsburgh, PA</i>)

ADDITIONAL INFORMATION

<i>Languages</i>	English (native), Gujarati (fluent), Hindi (fluent), French (conversational)
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Journal abbreviations used

JASTP | *Journal of Atmospheric and Solar Terrestrial Physics*
ApJ | *The Astrophysical Journal*

This CV was last updated on February 3, 2026.